VANSH SHARMA

AI/ML Engineer (MCA, AI & Data Science)

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PROFESSIONAL SUMMARY

- AI/ML Engineer (MCA, AI & Data Science) specializing in LLM fine-tuning, Retrieval-Augmented Generation, and multi-agent orchestration.
- Hands-on experience with parameter-efficient fine-tuning (QLoRA), LangChain/LangGraph agent systems, and production RAG pipelines built and evaluated end-to-end, from data preparation to deployment.
- Seeking an AI Engineer / GenAI Engineer role to build production-grade LLM systems.

SKILLS

Programming Languages: Python, C++, SQL

Frameworks & Libraries: LangChain, LangGraph, PyTorch, Hugging Face, Unsloth, Scikit-learn, XGBoost

AI / GenAI: RAG, Prompt Engineering, QLoRA Fine-Tuning, Multi-Agent Systems, GGUF Quantization, Ollama

Databases: FAISS, PostgreSQL, MySQL

Tools & Platforms: FastAPI, REST APIs, Streamlit, Docker basics, Git, GitHub, Jupyter, Google Colab

CS Fundamentals: OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

Graphic Era Hill University — Master of Computer Applications (AI & Data Science) *2024 – 2026*

- CGPA: 8.0 / 10

DIT University — Bachelor of Computer Applications *2021 – 2024*

- CGPA: 7.7 / 10

PROJECTS

Fine-Tuned LLM Job Information Extractor

Unsloth, QLoRA, PyTorch, Llama 3.2, Ollama, GGUF, FastAPI, json-repair

- Fine-tuned Llama-3.2-3B-Instruct using QLoRA (Unsloth) on a curated job-postings dataset to extract structured JSON entities (skills, experience, salary) from raw text.
- Reduced training loss from 2.32 to 0.13 quantized the model to 4-bit GGUF for low-latency local inference via Ollama.
- Implemented a json-repair validation layer, improving valid JSON extraction rate to >95% on a 50-sample held-out benchmark.
- Deployed an asynchronous FastAPI service handling real-time unstructured-to-structured schema extraction.

ResearchMind — Multi-Agent Research System

LangGraph, LangChain, Tavily API, Ollama, Llama 3.1, Streamlit

- Built a multi-agent iterative research pipeline in LangGraph orchestrating 4 specialized agents (Search, Scrape, Writer, Critic) with cyclic revision loops.
- Integrated Tavily API for real-time web retrieval, feeding structured context into agent prompts to reduce hallucinations.
- Migrated inference from proprietary APIs to a locally hosted Llama 3.1 (8B) model via Ollama, eliminating recurring per-token API costs.
- Developed an interactive Streamlit dashboard with state persistence, real-time execution logging, retry logic, and fallback handlers.

LexAI — Legal Document Intelligence Platform

LangChain, FAISS, Sentence-Transformers, Map-Reduce, Streamlit

- Developed an end-to-end legal RAG platform using FAISS and Sentence Transformers (all-MiniLM-L6-v2) for clause-level semantic search and question answering.
- Implemented a LangChain MapReduce summarization chain with token-aware chunking to synthesize 10,000+ word legal contracts exceeding typical LLM context windows.
- Integrated automated risk-scoring and clause-comparison modules; built a Streamlit UI for natural-language legal auditing.

CERTIFICATIONS

- Completed Google's 5-Day AI Agents Intensive Course (Kaggle) — agent architectures, agent sessions, agent-to-agent (A2A) communication
- Deloitte Australia — Data Analytics Job Simulation (2025)
- Participant, AI-Driven Solutions track, National-Level Hackathon